

Subset divisibility completely characterizes signed fair signatures

Research proof; two independent mathematical reviews completed

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Let $\mathcal{A}_{\mathbb{Q}}$ be the associative algebra with basis the empty word and all words of distinct letters from $X_1, \dots, X_n$. Multiplication is concatenation when the two letter sets are disjoint, and zero otherwise. Equivalently, it is the free associative algebra modulo the ideal generated by all words containing a repeated letter. Let $\mathcal{A}_{\mathbb{Z}}$ denote its integer-coefficient subring and let $\mathfrak{m}$ denote the positive-degree ideal. This ideal is nilpotent, so exponential and logarithm are finite polynomials over $\mathbb{Q}$.

A signed word is a product of the elements $1 + X_i = \exp(X_i)$ and their inverses $1 - X_i = \exp(-X_i)$. Negative letters are purely algebraic: they are not legal die faces.

Classical algebraic setting. This is the reduced free associative algebra used in Milnor's Magnus expansion of the reduced free group. The group generated by $1 + X_i$ identifies with that reduced free group by [1, Theorem 1.12 and Corollary 1.13]. The integral fixed-last-letter Lie basis used below is also classical; see [1, Lemma 6.15]. The elementary proofs are retained to make the application to the fair target self-contained. For positive words, equality of these reduced signatures is exactly strong M -equivalence in [2, Theorem 3.3]. No novelty is claimed for these algebraic tools.

Theorem 1. *For integers $c_1, \dots, c_n$, the fair signature*

$$T_c = \exp \left(\sum_i c_i X_i \right)$$

is represented by a signed word if and only if

$$|S|! \mid \prod_{i \in S} c_i \quad \text{for every nonempty } S \subseteq [n].$$

Thus integral membership in the reduced nilpotent group supplies no arithmetic obstruction beyond the subset-factorial divisibility already known for ordinary fair words. Any stronger obstruction for positive words must use positivity or some other semigroup constraint.

Lemma 2 (An integral multilinear Lie basis). *Fix a nonempty set of letters S and a distinguished $a \in S$. For each ordering π of $S \setminus \{a\}$, put*

$$B_\pi = [X_{\pi_1}, [X_{\pi_2}, \dots, [X_{\pi_{|S|-1}}, X_a] \dots]].$$

For $S = \{a\}$ use $B_\emptyset = X_a$. These form a basis of the multilinear Lie component on S over $\mathbb{Q}$. In this basis, the coefficient of B_π in a Lie element L is exactly the associative-word coefficient of $X_{\pi_1} \cdots X_{\pi_{|S|-1}} X_a$ in L . In particular, an integral-coefficient multilinear Lie element has integer coordinates in this basis.

Proof. Antisymmetry and the Jacobi identity rewrite every multilinear Lie bracket as a linear combination of brackets right-nested toward the fixed letter a . One can see this inductively from $[[U, V], W] = [U, [V, W]] - [V, [U, W]]$, always moving the bracket containing a to the right. Thus the displayed brackets span. When B_π is expanded associatively, its only word ending in X_a is $X_{\pi_1} \cdots X_{\pi_{|S|-1}} X_a$, with coefficient one. The ending- a coefficients therefore prove linear independence and the asserted coordinate formula. $\square$

Proof of Theorem 1. The coefficient of every word whose letter set is S in T_c is $\prod_{i \in S} c_i / |S|!$. Every signed word has integer coefficients, so the divisibility conditions are necessary.

For sufficiency, the conditions say $T_c \in \mathcal{A}_{\mathbb{Z}}$. We construct a signed word g equal to T_c by correcting degrees successively. Suppose $g^{-1}T_c$ agrees with 1 in degrees below k . Because g and g^{-1} have integer coefficients, the homogeneous part L_k of degree k of $g^{-1}T_c - 1$ has integer coefficients. It is also a Lie element: $\log g$ is a Lie polynomial by the finite Baker–Campbell–Hausdorff formula, $\log T_c = \sum c_i X_i$ is a Lie polynomial, and hence $\log(g^{-1}T_c)$ is one too. Its lowest nonzero homogeneous part is L_k .

Decompose L_k by its letter sets. Lemma 2 expresses each component as an integer combination of the B_π . Each factor $1 + B_\pi$ is itself represented by a signed word, obtained by the identically nested group commutator. In fact if U, V have disjoint fixed supports and positive degrees, then

$$(1 + U)(1 + V)(1 - U)(1 - V) = 1 + [U, V]$$

whenever $U^2 = V^2 = UVU = VUV = 0$, as holds for such fixed-support elements in this algebra. Induction gives the exact assertion for B_π . Also $(1 + B_\pi)^b = 1 + bB_\pi$ for every integer b , because $B_\pi^2 = 0$.

Multiply g on the right by the product of these factors with their integer coordinates. This product is $1 + L_k$ modulo degrees greater than k , so the new $g^{-1}T_c$ agrees with 1 through degree k . Starting with $g = 1$ and continuing through degree n constructs the required signed word. $\square$

Equal counts. For $c_i = M > 0$, the divisibility conditions are equivalent to every prime $p \leq n$ dividing M . Thus the primorial $M = \prod_{p \leq n} p$ always has an integral signed fair signature. In particular $n = 4, M = 6$ and $n = 5, M = 30$ pass all reduced-group integrality tests. This does not assert that ordinary fair dice with these parameters exist.

Exact finite checks. The companion script `check_signed_signatures.py` performs the entire degree-correction algorithm in the integer algebra, verifies the homogeneous Lie-basis identities at every step, and compares all final coefficients to T_c . It records compact commutator coordinates rather than expanding the possibly long signed word.

References

- [1] Jacques Darné, *Milnor invariants of braids and welded braids up to homotopy*, Algebraic & Geometric Topology 24 (2024), 1277–1320, doi:10.2140/agt.2024.24.1277. The referenced theorem numbering was checked in the author’s arXiv:1904.10677v2.
- [2] Wen Chean Teh, *Parikh matrices and strong M -equivalence*, International Journal of Foundations of Computer Science 27 (2016), 545–556, doi:10.1142/S0129054116500155; arXiv:1501.07354.